



1. A thief after stealing, went off in a Santro with a speed 40 kmph. The owner of the house discovered the theft after half an hour and then he set off on his wife at a speed of 50 kmph. At what time, he will catch up the thief after his start: **(TSD) NIMCET-2007**
(a) 2 hrs. (b) 2.30 hrs.
(c) 3 hrs. (d) none
2. Two trains each of length 500 m are travelling in opposite direction with speeds 45 kmph and 30 kmph respectively. In what time, slower train will pass the driver of faster train. **(TSD) NIMCET-2007**
(a) 30 sec (b) 60 sec (c) 50 sec (d) none
3. A car is filled with $4\frac{1}{2}$ L of fuel for a around trip. If amount of fuel taken while goings is $\frac{1}{4}$ th more than the amount taken for coming, what is the amount of fuel consumed when coming back? **(TSD) NIMCET-2008**
(a) 1.5 (b) 2 (c) 1.75 (d) None
4. If a man walks at the rate of 4 km/h, he misses a train by only 6 min. However, if he walks at the rate of 5 km/h he reaches the station 6 min before the arrival of the train. Find the distance covered by him to reach the station. **(TSD) NIMCET-2008**
(a) 4 km (b) 7 km (c) 9 km (d) 5 km
5. A person travels on a cycle from home to church on a straight road with wind against him. He took 4 h to reach there. On the way back to the home, he took 3 h to reach as wind was in the same direction. If there is no wind, how much times does he take to travel from home to church? **(TSD) NIMCET-2008**
(a) 3 h 35 min 12 s (b) 3 h 32 min 32 s
(c) 3 h 30 min 00 s (d) 3 h 25 min 42 s
6. A train after travelling 60 km meets with an accident and then proceeds at $\frac{3}{4}$ of its former rate and arrives at the terminus 40 minutes late. Had the accident happened 25km further on, it would have arrived 10 minutes sooner. Find the speed of the train and the distance, respectively. **(TSD) NIMCET-2009**
(a) 160 km / hr, 150 km (b) 160 km / hr, 140 km
(c) 50 km / hr, 160 km (d) 40 km / hr, 160 km
7. A culprit was spotted by the police from a distance of 250 m. When the police men started running towards the culprit at a speed of 10 km/h, the culprit also fled. If his speed was 8 km/h, find out how far the culprit had run before he was overpowered. **(TSD) NIMCET-2010**
(a) 2 km (b) 1 km
(c) 1.5 km (d) 0.8 km
8. A doctor said to his compounder "I go to see the patients at their residence after every 3:30 hours. I have already gone to the patient 1:20 hours ago and next time I shall go at 1.40 pm". At what time this information was given to the compounder by the doctor ? **(TSD) NIMCET-2011**
(a) 10.10 a.m. (b) 11.30 a.m.
(c) 11.20 a.m. (d) None of these
9. If a man walks at the rate of 4 km/h ,he misses a train by only 6 min. However, if he walks at the rate of 5km/h he reaches the station 6 min before the arrival of the train. The distance covered by him to reach the station is **(TSD) NIMCET-2012**
(a) 4 km
(b) 7 km
(c) 9 km
(d) 5 km
(e)
10. A train takes 18 s to pass completely through a station 162 m long and 15 s through another station 120 m long, at the same speed. What is then length of the train in metre? **(TSD) NIMCET 2013**
(a) 70 (b) 80 (c) 90 (d) 100
11. A doctor said to his compunder 'I go to see the patients at their residence after every 3 h 30 min. I have already gone to the patient 1 h 20 min ago and next time I shall go at 1:40 pm. At what time this information was given to the compounder by the doctor?**(TSD) NIMCET 2013**
(a) 11:30am (b) 11:20 am
(c) 10:10 am (d) None of these
12. Karan and Arjun run a 100 m race,where Karan beats Arjun by 10 m. To do a favour of Arjun, Karan starts 10 m behind the starting line in a second 100 m race. They both run at their earlier speeds. Which of the following is true in connection with the second race?
(a)Karan and Arjun reach the finishing line simultaneously **(TSD) NIMCET 2013**
(b)Karan beats Arjun by 1 m
(c)Arjun beats Karan by 11 m
(d)Arjun beats Karan by 1 m
13. A runs $1\frac{2}{3}$ times as fast as B. If A gives B start of 80m, how far must the winning post be so that A and B might reach it at the same time? **(TSD) NIMCET-2018**
(a) 200m (b) 300m
(c) 400m (d) 160m





14. A body travels from A to B in 10 s with a speed of 50 km/h and returns with a speed of 100km/h in 5 s. The average speed and the average and the average velocity for the whole journey is

(TSD) NIMCET-2018

- (a) $17.5 \text{ ms}^{-1}, 0 \text{ km/h}$ (b) $16.5 \text{ ms}^{-1}, 0 \text{ km/h}$
(c) $15.5 \text{ ms}^{-1}, 0 \text{ km/h}$ (d) $18.5 \text{ ms}^{-1}, 0 \text{ km/h}$

15. Navjivan Express from Ahmedabad to Chennai leaves Ahmedabad at 6:30 am and travels at 50 kmph towards Baroda situated 100 km away. At 7:00 am Howrah-Ahmedabad and travels at 40 kmph. At 7:00 am Mr. Shah, the traffic controller at Baroda realizes that both train are running on the same track. How much time does he have to avert a head- on collision between the two train?

- (a) 15 min (b) 20 min
(c) 25 min (d) 30 min

(TSD) NIMCET-2019

16. A train overtakes two persons who are walking in the same direction in which the train is going, at the rate of 2 kmph and 4 kmph and passes them completely in 9 sec and 10 sec respectively. The length of the train is **(TSD) NIMCET-2019**

- (a) 72 m (b) 54 m (c) 50 m (d) 45 m

17. A runs $1\frac{2}{3}$ times as fast as B. If A gives B a start of 80m, how far must winning post be so that A and B might reach it at the same time? **(TSD) NIMCET-2020**

- (a) 200m (b) 400m (c) 300m (d) 160m

18. Mr. Kumar drives to work at an average speed of 48 km/hr. the time taken to cover the first 60% of the distance is 10 min more than the time taken to cover the remaining distance. How far is his office ? **(TSD) NIMCET-2020**

- (a) 30km (b) 40km (c) 45km (d) 48km

19. If a man walks at the rate of 4km/hr, he misses a train by only 6 min. However, if he walks at the rate of 5km/hr, he reaches the station 6 min before the arrival of the train. The distance covered by him reach the station is **(TSD) NIMCET-2020**

- (a) 4 km (b) 7 km (c) 9 km (d) 5 km

20. Rajdhani train running at a speed of 54 km/h crosses a platform of length same as that of the train in 36 sec. If a duronto train, which is 230 m long crosses the same platform in 25 sec, then find speed of duronto train (in km/h)? **(TSD) NIMCET-2023**

- (a) 54 km/h (b) 72 km/h (c) 84 km/h (d) 90 km/h

21. Two cars, Car A and Car B, are travelling on a highway. Car A starts from point X and travels at a constant speed of 60 km/h, while Car B starts from the same point X but travels at a constant speed of 80 km/hr. If both cars travel for 1.5 hours, what is the difference in distance covered by Car B compared to Car A? **(TSD) NIMCET-2024**

- (a) 20 km (b) 25 km
(c) 35 km (d) 30 km

22. Ramu visits Delhi on every 15 days and Samu goes to Delhi every 20 days. They met at Delhi 5 days back. After how many days, from today, they will meet at Delhi next time?

(TSD) NIMCET-2024

- (a) 35 (b) 55
(c) 60 (d) 65

ANSWER

1	a	2	d	3	b	4	a	5	d
6	c	7	b	8	b	9	a	10	c
11	a	12	d	13	a	14	d	15	b
16	c	17	a	18	b	19	a	20	b
21	d	22	b						

SOLUTION

1. (a) Speed of thief = 40 km/hr
Speed of owner = 50 km/hr
Distance = Speed x time
 $40 \times t = 50 (t - 1/2)$
 $4/5 t = 1/2$
 $T = 5/2 = 2.5 \text{ hr}$
2.30 hr

2. (d)
Relative speed = $(45 + 30) \text{ km/hr}$
 $= 75 \times \frac{5}{18} \text{ m/sec}$
 $= \frac{125}{6} \text{ m/sec}$
Distance covered = Length of the slower train
Distance covered = 500
Required time = $500 \times \frac{6}{125} = 24 \text{ sec}$.

3. (b) If x is the amount of fuel consumed while coming back.
Then $x + x + \frac{x}{4} = 4\frac{1}{2}$
 $\Rightarrow \frac{9x}{4} = \frac{9}{2}$
 $x = 2$

4. (a)
If d is the distance covered, then if 't' hour is the required time to catch the train in time, then

$$\frac{d}{4} = t + \frac{1}{10}$$

and $\frac{d}{5} = t = \frac{1}{10}$
 $\Rightarrow \frac{d}{20} = \frac{1}{5}$
 $\Rightarrow d = \frac{20}{5} = 4 \text{ km}$

5. (d)
Let the speed of person and that of wind be u and v km/h respectively. d

Then $\frac{d}{u-v} = 4$
and $\frac{d}{u+v} = 3$
 $\Rightarrow 4(u-v) = 3(u+v)$
 $\Rightarrow u = 7v$
 $\Rightarrow \frac{d}{u+\frac{u}{7}} = 3$
 $\Rightarrow \frac{d}{(\frac{8u}{7})} = 3$
 $\Rightarrow \frac{d}{u} = \frac{24}{7}$





Hence, required time = $\frac{24}{7}$ h = 3 h 25 min 42 sec

6. (c) Let x km/h be the speed of the train and y km be the distance.

$$\text{Then, } \frac{y-60}{\frac{3}{4}x} = \frac{25}{x} + \frac{y-85}{\frac{3}{4}x} + \frac{10}{60}$$

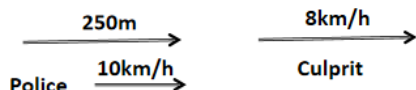
$$\Rightarrow \frac{25}{\frac{3}{4}x} = \frac{25}{x} + \frac{1}{6}$$

$$\Rightarrow \frac{25}{3x} = \frac{1}{6}$$

$$\Rightarrow x = 50 \text{ km/h}$$

$$\text{and } y = 160 \text{ km}$$

7. (b)



Police men running speed = 10 km/h

Culprit speed = 8 km/h

Police speed - Culprit speed

$$10 - 8 = 2 \text{ km/h}$$

Distance = 250m

Speed = 2 km/h

$$= 2 \times \frac{5}{18} \text{ m/sec.}$$

[distance = speed x time]

Time = distance/ speed

$$\text{Time} = \frac{250}{2 \times 5}$$

Time = 450 sec.

Distance = Culprit speed x time

$$= 8 \text{ km/h} \times 450 \text{ sec.}$$

$$= 8 \times \frac{5}{18} \times 450$$

$$= 4 \times 250 = 1000 \text{ m}$$

8. (b) Doctor go to see the patients before

$$1:40 \text{ pm} = 1:40 \text{ pm} - 3:30$$

$$= 13:40 \text{ pm} - 3:30 \text{ h} = 10:10 \text{ am}$$

$$\text{Hence, required time} = 10:10 + 1:20 = 11:30 \text{ am}$$

9. (a) Let the distance covered by him is x km, then by condition,

$$\frac{x}{4} - \frac{x}{5} = \frac{12}{60}$$

$$\Rightarrow \frac{x}{20} = \frac{1}{5}$$

$$\therefore x = 4 \text{ km}$$

10. (c) Let the length of the train be x m.

$$\text{Then, } \frac{x+162}{18} = \frac{x+120}{15}$$

$$\Rightarrow \frac{x+162}{6} = \frac{x+120}{5}$$

$$\Rightarrow 6x + 720 = 5x + 810$$

$$\Rightarrow x = 90 \text{ m}$$

11. (a) Doctor's previous time to see the patients

$$= 1:40 \text{ pm} - 3 \text{ h } 30 \text{ m} = 10:10 \text{ am}$$

So, the information given to the compounder by the doctor

$$= 10:10 \text{ am} + 1:20 = 11:30 \text{ am}$$

12. (d) In 1st 100 m race, distance covered by Karan = 100 m

and distance covered by Arjun = 90 m

In 2nd 100 m race, distance covered by Karan = 110 m

and distance covered by Arjun = ?

100 m of Karan = 90 m of Arjun

$$110 \text{ m of Karan} = \frac{90 \times 110}{100} = 99 \text{ m of Arjun}$$

$\therefore$ Karan beats Arjun by 1 m.

13. (a) A runs $1\frac{2}{3}$ times as fast as B.

Hence, ratio of speed of A and B = 5:3 or we can say that A moves 5 m in the same time in which B moves 3 m.

Hence, A gains 2 m over B by running 5 m.

$$\therefore \text{Required distance} = \frac{5}{2} \times 80 = 200 \text{ m}$$

14. (d) Speed while travelling from A to B = 50 km/h

Speed while travelling from B to A = 100 km/h

$$\therefore \text{Required average speed} = \frac{2 \times 50 \times 100}{50 + 100} \times \frac{1000}{3600} = 18.5 \text{ m/s}$$

Now, average velocity is 0 km/h as displacement is zero.

15. (b) Initially, 6:30 AM distance between both trains is 100 km now at

7:00 AM, Navjeevan travels 25 km, so remaining distance

between both trains is 75 km. Again at 7:30 AM, Navjeevan

travels is 25 km more and also Howrah travel 20 km.

So, final distance between them is 75 - (25 + 20) km 30 km, so

time taken to travel 30 km at relative speed (50 km/h + 40 km/h)

90 km/h 30 h

$$= \frac{30}{90} \text{ h} = \frac{1}{3} \times 60 = 20 \text{ min}$$

16. (c) Let the length of the train be x m and speed by y m/s,

Speed of the first person = 2 km/h

$$= 2 \times \frac{5}{18} = \frac{5}{9} \text{ m/s}$$

Speed of the second person = 4 km/h

$$= 4 \times \frac{5}{18} = \frac{10}{9} \text{ m/s}$$

When pass first person, $\frac{x}{y - \frac{5}{9}} = 9$

$$\Rightarrow 9y - 5x = x$$

$$\Rightarrow 90y - 50 = 10x$$

...(i)

and when pass second person $\frac{x}{y - \frac{10}{9}} = 10$

$$\Rightarrow 90y - 100 = 9x$$

...(ii)

Subtracting Eqs. (i) and (ii), we get

$$90y - 50 - 90y + 100 = 10x - 9x$$

$$\Rightarrow x = 50$$

So, the length of the train 50 m

17. (a) We have,

$$\text{Speed of A : Speed of B} = \frac{5}{3} : 1 = 5 : 3$$

$\Rightarrow$ In a race of 5 meters, A runs 5 meters and B runs 3 meters.

$\Rightarrow$ A gains 80 metres over B in a race of $\frac{5}{2} \times 80 = 200 \text{ m}$

18. (b) Let the total distance be x km.

Time taken to cover remaining 40% of x distance is

$$t_1 = \frac{40 \times x}{100 \times 48}$$

Since, time taken to cover first 60% of x distance is $t_2 = t_1 + \frac{10}{60}$ hrs.

$$\Rightarrow t_2 = \frac{60 \times x}{100 \times 48}$$



$$\begin{aligned}\therefore \frac{60 \times x}{100 \times 48} &= \frac{40 \times x}{100 \times 48} + \frac{10}{60} \\ \Rightarrow \frac{60 \times x}{100 \times 48} &= \frac{40 \times x}{100 \times 48} + \frac{10}{60} \\ \Rightarrow \frac{100 \times 48}{20x} &= \frac{10}{60} \\ \Rightarrow \frac{4800}{20x} &= \frac{10}{60} \\ \Rightarrow x &= \frac{10}{60} \times \frac{4800}{20} \\ \Rightarrow x &= 40 \text{ km.}\end{aligned}$$

19. (a) Let the correct time of train arrival be t and distance to reach the station be d .

Distance = Speed $\times$ Time

$$1 \text{ min} = \frac{1}{60} \text{ h}$$

$$d = \frac{4 \times (t+6)}{60}$$

$$d = \frac{5x(t-6)}{60}$$

From Eqs. (i) and (ii)

$$\frac{4 \times (t+6)}{60} = \frac{5 \times (t-6)}{60}$$

$$5t - 4t = 30 + 24$$

$$t = 54 \text{ min}$$

Putting the value of 't' in Eq. (i)

$$d = \frac{4 \times (54+6)}{60}$$

$$d = 4 \times \frac{60}{60}$$

$$d = 4 \text{ km.}$$

$\therefore$ The distance covered by him to reach the station is 4 km.

$\therefore$ The correct answer in option (a).

20. (b) Let length of Rajdhani train be x .

Then, the length of platform is also x .

Speed of train = 54 km/h

$$= 54 \times \frac{5}{18} \text{ m/s}$$

$$= 15 \text{ m/s}$$

According to the question,

$$= \frac{(x+x)}{15} = 36$$

$$\Rightarrow 2x = 540$$

$$x = 270 \text{ m}$$

A Duronto train of length 230 m crosses the platform in 25s.

Let the speed of train be V m/s.

$$\Rightarrow \frac{270+230}{V} = 25$$

$$\Rightarrow V = 20 \text{ m/s}$$

Speed of train = 20 m/s

$$= 20 \times \frac{18}{5}$$

$$= 72 \text{ km/h}$$

21. (d)

A $\rightarrow$ 60 km/h $\rightarrow$ 90 km

B $\rightarrow$ 80 km/h $\rightarrow$ 120 km

$$120 - 90 = 30$$

So option (d) is correct.

22. (b)

$$\begin{array}{cc} R & S \\ 15 & 20 \end{array}$$

60 days
55 days

5 Days